Rationalise Complex Numbers

Key Points

A complex number, $z = x + yi$ has a complex **conjugate**

$$z^* = x - yi$$

To divide complex numbers we **rationalise** by multiplying by the complex conjugate over itself

$$\frac{w}{z} = \frac{w}{z} \times \frac{z^*}{z^*} = \frac{w \times z^*}{z \times z^*}$$

This causes the bottom to only be real as

$$zz^* = x^2 + y^2$$

Basic Skills

Q1. Write down the complex conjugate of the following

a) $z = 2 + 3i$ **b)** $z = 4 - i$ **c)** $z = 2i - 5$

Q2. For $z = 3 + 5i, w = 5 - 2i$ and $v = 7i - 9$ calculate the following

a) $z \times w$

b) $z \times v$

c) $w \times v^*$

d) $z \times z^*$

Q3. What do we multiple $\frac{3+2i}{4+5i}$ by to rationalise it

Competency Skills

Q4. Rationalise the following complex numbers

a) $\frac{3-2i}{2+2i}$ **b)** $\frac{6+5i}{3-2i}$ **c)** $\frac{5(i-6)}{i-1}$ **e)** $\frac{3i-2}{(3-2i)^2}$ **f)** $\frac{(2-i)(i-2)}{5-2i}$

Q5. Solve the equation $(3 - 5i)z = 10 + 3i$

Q6. Find real numbers a and b such that $\frac{1-6i}{3+2i} = \frac{a+bi}{2+5i}$

Q7. Solve the equation $(4 + 9i)(z - 3i + 2) = 2 - 5i$

Q8. In general for $z = x + yi$ calculate $z + z^*$ and $z - z^*$ as an expression involving x and y

Advanced Skills

Q9. Explain how we divide a complex number z by a complex number w

Q10. In general for $z = x + yi$ calculate $\frac{1}{z} + \frac{1}{z^*}$ and $\frac{1}{z} - \frac{1}{z^*}$ as an expression involving x and y

Q11. Find real numbers a and b such that $\frac{a}{2-i} + \frac{bi}{5+2i} = 2 + 3i$

Q12. Solve the simultaneous equations $z + 2iw = 1$, $2z - 8w = 22$

Q13. Deduce the inverse of a complex number $z = x + yi$ and state how this relates to rationalising the denominator when dividing by z

Extension

Q14. Can you write a general formula for rationalising $\frac{a+bi}{c+di}$ in terms of a, b, c and d